MAL3018

COMPUTING PROJECT

PROJECT PROPOSAL:
ForensicX: Unified Analysis Framework

Prepared by,

Azizul Hakim Hassan
BSCS 2409162
BSC (HONS) Computer Science
(Cyber Security)
Year 3 Semester 1
bscs2409162@peninsulamalaysia.edu.my
azizul.hassan@students.plymouth.ac.uk

Table of Contents

Background and Problem Statement 3

Objectives 3

Scope of the Project 4

Literature Review / Related Works 5

4.1 Summary of Related Systems and Tools 5

4.2 Identified Gaps 6

4.3 How ForensicX Addresses These Gaps 6

Methodology 7

5.1 Project Approach 7

5.2 Tools and Technologies 7

5.3 System Design / Architecture 8

5.4 Implementation Plan 9

5.5 Testing and Evaluation 10

Expected Deliverables 10

Risk Plan 12

Summary of Mitigation Approach 13

Project Timeline 14

Milestone Overview 14

Resources Required 16

8.1 Hardware Resources 16

8.2 Software Resources 16

8.3 Online Resources and APIs 17

8.4 Human and Support Resources 17

Expected Outcome and Significance 18

9.1 Expected Outcomes 18

9.2 Significance of the Project 18

9.3 Long-Term Vision 19

References 20

Background and Problem Statement

Ransomware and cloud-targeted attacks have become one of the most critical cybersecurity challenges in recent years. These incidents cause severe operational and financial disruptions, often encrypting vital data, interrupting business continuity, and exploiting cloud storage environments. Digital forensic investigations into such attacks require collecting and correlating evidence from diverse sources including volatile memory, file systems, system logs, and cloud metadata.

However, current forensic practices remain **fragmented and tool dependent**. Memory analysis, log correlation, and cloud investigations often rely on separate tools, formats, and manual workflows. This fragmentation slows down investigations, introduces inconsistencies in the chain of custody, and makes evidence less reliable in legal or audit contexts.

To address these challenges, **ForensicX** aims to provide a **unified analysis framework** that integrates evidence from multiple sources into a single, coherent environment. The framework will automate artifact correlation, detect ransomware indicators, and generate reproducible forensic reports. By merging memory, log, and cloud data within one workflow, ForensicX enhances the speed, accuracy, and reliability of ransomware and cloud attack investigations.

Objectives

The **ForensicX: Unified Analysis Framework** project aims to streamline digital forensic investigations for ransomware and cloud-based attacks through automation and evidence integration. The key objectives are:

1. **To develop** a unified forensic framework that integrates evidence from multiple sources including memory, system logs, and cloud metadata into a single automated analysis workflow.
2. **To implement** ransomware detection and correlation mechanisms that identify key artifacts, reconstruct attack timelines, and present findings coherently.
3. **To evaluate** the framework's effectiveness using simulated attack scenarios in a controlled virtual lab, measuring accuracy, reliability, and investigation speed.

Scope of the Project

The scope of this project defines the boundaries of what **ForensicX** will and will not cover to ensure a focused and achievable outcome.

In Scope	Out of Scope
Development of a modular forensic framework that integrates memory, log, and cloud evidence sources.	Deployment or testing in live enterprise environments.
Implementation of ransomware artifact detection using entropy analysis, YARA signatures, and file behavior rules.	Use of real or live ransomware samples in production systems.
Correlation of forensic artifacts to reconstruct incident timelines and detect relationships between evidence sources.	Full-scale SIEM integration or commercial-grade deployment.
Design of a web-based dashboard and automated report generator (PDF/HTML) for investigation summaries.	Advanced machine learning or AI-driven forensic prediction models.
Evaluation of the system in a controlled virtual lab environment with synthetic ransomware and cloud test data.	Real-world law enforcement case investigations.

Literature Review / Related Works

4.1 Summary of Related Systems and Tools

Tool / System	Description	Strengths	Limitations	Reference
Volatility (Volatility 3)	Memory forensics framework for analyzing RAM dumps to extract processes, DLLs, and network data.	Powerful memory analysis active plugin support widely used in research.	Limited to memory artifacts lacks log and cloud integration.	Volatility Foundation (2021)
YARA	Rule-based tool for detecting malware patterns and suspicious files.	Flexible syntax effective for malware artifact identification.	High maintenance of rules heuristic-based, prone to false positives.	VirusTotal (2020)
GRR Rapid Response	Remote live forensics and incident response tool by Google.	Scalable endpoint collection effective remote triage.	Not ransomware-specific lacks unified reporting and cross-source correlation.	Cohen et al. (2011)
Velociraptor	Open-source endpoint visibility and forensic collection platform.	Strong artifact collection flexible query language.	Steep learning curve complex setup limited cloud integration.	Velociraptor (2023)
Timesketch	Open-source timeline analysis and visualization tool.	Excellent collaborative timeline visualization.	Requires separate ingestion	Google (2019)

			lacks memory and cloud support.	
Magnet AXIOM (Commercial)	Proprietary all-in-one forensic suite covering disk, memory, mobile, and cloud.	Comprehensive features enterprise-grade reporting.	Closed-source Expensive not suited for academic reproducibility.	Magnet Forensics (2017)
Elastic Security (ELK Stack)	Log ingestion, correlation, and visualization platform.	Scalable powerful search and visualization.	Focused on detection and monitoring, not forensic preservation.	Elastic NV (2021)

4.2 Identified Gaps

- No open-source framework currently **integrates memory, logs, and cloud metadata** into one workflow.
- Existing tools **lack automated cross-source correlation** and **chain-of-custody reporting**.
- Commercial solutions (e.g., Magnet AXIOM) are powerful but **not reproducible** for academic use.

4.3 How ForensicX Addresses These Gaps

Gap	**ForensicX Solution**
Fragmented forensic tools and data sources	Unified pipeline integrating memory, logs, and cloud evidence.
Lack of automated cross-correlation	Built-in correlation engine reconstructing full attack timelines.
Missing standardized reporting	Automated PDF/HTML report generation with IOCs and evidence hashes.
Limited reproducibility in commercial tools	Fully open, Python-based modular prototype for research and teaching.

Methodology

5.1 Project Approach

The project adopts an **Agile development approach**, allowing iterative testing and improvement of each forensic module. This approach ensures that each phase from data collection to reporting can be continuously refined based on test results and supervisor feedback.

Each sprint will deliver a working component:

- **Sprint 1–2:** Core architecture setup and memory forensics module.
- **Sprint 3–4:** Log parser and correlation engine integration.
- **Sprint 5–6:** Cloud forensic collector and dashboard/report generator.
- **Sprint 7:** Testing, evaluation, and refinement.

5.2 Tools and Technologies

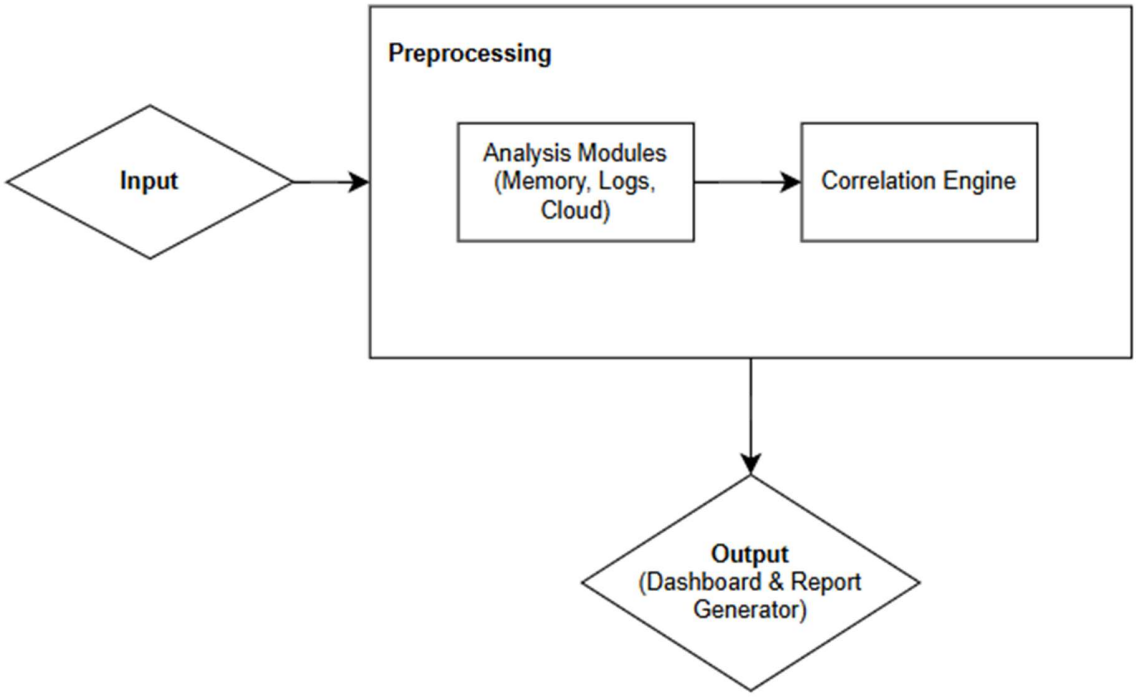
Category	Tools / Technologies	Purpose / Function
Programming Language	Python 3.x	Main development language for modules and framework integration.
Memory Forensics	Volatility 3	Extraction of RAM artifacts such as processes, DLLs, and network connections.
Ransomware Detection	YARA Rules, Entropy Calculation	Identify encrypted files, detect abnormal patterns and indicators of compromise (IOCs).
Log Collection & Correlation	Windows Event Logs (EVTX), Sysmon, Python parsers	Extract and correlate logs related to ransomware activity.
Cloud Metadata Analysis	Google Drive API, AWS S3 API, Microsoft Graph API	Collect cloud file activity metadata for forensic analysis.
Database / Data Storage	SQLite or Elasticsearch	Store extracted artifacts and correlation results.
Dashboard & Visualization	Streamlit / Flask	Interactive dashboard for visualization and reporting.

Reporting	ReportLab / Markdown-PDF	Automated generation of forensic reports (PDF/HTML).
Testing Environment	VMware Workstation 16, Ubuntu Forensic Server, Windows Client VM	Controlled lab environment for safe simulation and analysis.

5.3 System Design / Architecture

The ForensicX system follows modular architecture, consisting of several key components that operate in sequence to ensure comprehensive forensic analysis.

System Workflow:



Component	Description
Memory Forensics Module	Analyzes RAM dumps using Volatility and YARA to detect suspicious processes and file behaviors.
Log Parser & Correlation Engine	Parses Windows Event Logs and Sysmon data, then correlates system, network, and process events.
Cloud Forensic Collector	Retrieves cloud activity metadata (file uploads, deletions, access logs) via APIs for evidence correlation.
Ransomware Artifact Detector	Uses entropy analysis and pattern matching to identify encrypted or tampered files.
Dashboard & Report Generator	Combines all correlated evidence into a unified timeline and generates investigator-ready PDF/HTML reports.

5.4 Implementation Plan

Phase	Task	Expected Output
Phase 1	Framework setup, Volatility & YARA integration	Working memory forensics module
Phase 2	Log parser and correlation engine development	Log analysis + unified correlation
Phase 3	Cloud forensic collector implementation	Cloud metadata integration
Phase 4	Dashboard & report generator	Functional visualization and reporting
Phase 5	Testing & validation	Evaluation report and finalized prototype

5.5 Testing and Evaluation

Testing will be conducted in an **isolated virtual lab** to ensure ethical and safe experimentation.

Aspect	Description
Environment	Ubuntu forensic server and Windows VM with synthetic ransomware samples.
Test Scenarios	Simulated ransomware encryption, cloud metadata tampering, log injection.
Metrics	Detection accuracy, correlation completeness, report consistency.
Validation	Compare expected vs. detected indicators (file hashes, timestamps, process chains).
Outcome	Verified prototype demonstrating cross-source forensic integration.

Expected Deliverables

The **ForensicX** project will produce both software and documentation deliverables that demonstrate the framework's capabilities, technical implementation, and evaluation results.

Category	Deliverable	Description / Expected Output
Software System	**ForensicX Prototype Framework**	A Python-based modular system integrating memory, log, and cloud forensic modules with automated correlation and reporting.
Analysis Modules	**Memory Forensics, Log Parser, Cloud Collector**	Individual functional modules for artifact extraction and evidence analysis.
Correlation Engine	**Cross-Source Evidence Correlator**	Component that reconstructs attack timelines by linking artifacts across memory, logs, and cloud metadata.

Report Generator	**PDF/HTML Report Automation Tool**	Automatically generates standardized forensic reports including file hashes, timestamps, IOCs, and chain-of-custody information.
Dashboard Interface	**Streamlit / Flask Dashboard**	Interactive dashboard displaying results, evidence summaries, and timeline visualization.
Test Environment Setup	**Virtual Lab Environment (VMs)**	Configured Ubuntu forensic server and Windows VM for controlled testing with synthetic ransomware artifacts.
Documentation	**Technical Report & User Guide**	Comprehensive project documentation covering methodology, architecture, testing, and user instructions.
Presentation Materials	**Final Presentation Slides & Demonstration**	Visual and verbal presentation summarizing project goals, design, outcomes, and evaluation results.

These deliverables will collectively demonstrate the system's ability to unify fragmented forensic processes into a single, automated workflow. The outputs will be reproducible, open-source, and suitable for both academic evaluation and future research extension.

Risk Plan

The development of **ForensicX** involves both technical and operational risks. The following table identifies potential risks, their likelihood, impact, and corresponding mitigation strategies to ensure project success.

Risk Category	Description of Risk	Likelihood	Impact	Mitigation Strategy
Technical Risk	Integration challenges between memory, log, and cloud modules due to differing data formats or APIs.	Medium	High	Conduct early prototype integration and modular testing after each phase. Use standardized data structures (JSON/CSV).
Tool Compatibility	Dependency or version conflicts in Volatility, YARA, or API libraries.	Medium	Medium	Use virtual environments and document dependency versions. Regularly update and validate compatibility.
Data Risk	Accidental corruption or loss of test forensic data.	Low	High	Maintain frequent backups of project files and test datasets on external drives. Use version control (Git).
Security Risk	Exposure to real ransomware samples or unsafe malware during testing.	Low	High	Use only synthetic or sanitized test files. Run all experiments in isolated virtual machines (offline).
Time Management Risk	Overlapping academic workload or unforeseen delays in module development.	Medium	Medium	Follow Agile iterations; prioritize core components before optional features. Maintain weekly progress tracking.

Performance Risk	System performance may degrade when processing large forensic datasets.	Medium	Medium	Optimize correlation algorithms; limit dataset size during prototype testing. Use indexing or caching where possible.
Knowledge / Skill Gap	Limited prior experience with certain forensic APIs or libraries.	Medium	Medium	Conduct preliminary research, use online documentation, and seek supervisor guidance.
Ethical / Legal Risk	Misuse of forensic tools or unintentional breach of privacy laws during testing.	Low	High	Ensure compliance with ethical guidelines; only use self-generated or open test data. Obtain supervisor approval before testing.

Summary of Mitigation Approach

The project adopts a **preventive risk management strategy**, where each identified risk is monitored through regular progress reviews. Key preventive actions include:

- Developing and testing modules incrementally.
- Maintaining an offline, isolated test environment.
- Performing regular data backups and version control.
- Prioritizing essential functionality before advanced features.

This approach ensures that technical, ethical, and time-related risks are minimized, and the project remains on track for completion.

Project Timeline

The project will be completed over a **20-week period** (approximately 5 months), divided into five major phases: planning, development, integration, testing, and reporting. The timeline below outlines key tasks, milestones, and expected outputs.

Weeks	Task / Milestone	Expected Output
1–2	Literature review, finalize project requirements	Problem statement, requirements specification document
3–4	Lab setup (Ubuntu forensic server, Windows VM, Sysmon, dependencies)	Configured test environment
5–8	Develop memory forensics and ransomware detection modules (Volatility + YARA integration)	Working prototype of memory analysis module
9–12	Implement log parser and correlation engine	Functional cross-source correlation between logs and memory artifacts
13–16	Develop cloud forensic collectors and integrate dashboard/report generator	Unified analysis framework with reporting capability
17–18	System integration and refinement; debugging	Stable end-to-end prototype ready for evaluation
19	Testing and evaluation using synthetic ransomware scenarios	Evaluation results and system performance metrics
20	Documentation, report writing, and final presentation	Final report submission and project presentation

Milestone Overview

Phase	Focus Area	Milestone
Phase 1	Research & Planning	Requirements, architecture, and literature review completed
Phase 2	Module Development	Core forensic modules (memory, logs) functional

Phase 3	System Integration	Cloud collector and dashboard integrated
Phase 4	Evaluation	Framework validated in lab environment
Phase 5	Reporting & Presentation	Final report and presentation completed

Resources Required

The successful implementation of **ForensicX: Unified Analysis Framework** requires a combination of hardware, software, and online resources to support development, testing, and evaluation in a secure and reproducible environment.

8.1 Hardware Resources

Item	Specification / Description	Purpose
Personal Laptop / Workstation	Intel Core i5-11400H CPU, 16GB RAM, 1TB SSD, NVIDIA RTX 3050 Ti	Primary development and testing machine
Virtual Machines (VMs)	Ubuntu Server (Forensic Host) and Windows 10 Client	Controlled lab setup for isolated ransomware and forensic testing
External Storage (USB Drive / SSD)	32GB or larger	Offline storage for evidence transfer and backup
Network Setup	Local network between VMs (no internet connection)	Safe internal communication and log collection testing

8.2 Software Resources

Category	Software / Tool	Purpose / Function
Operating Systems	Ubuntu Server, Windows 10	Forensic environment and client simulation
Programming Tools	Python 3.x, Visual Studio Code	Development and debugging of framework modules
Forensic Tools	Volatility 3, YARA, Sysmon, FTK Imager	Memory forensics, pattern detection, and evidence extraction
Data Storage	SQLite / Elasticsearch	Storing correlated forensic data and search queries
Visualization & Dashboard	Streamlit / Flask	Web-based dashboard for visualization and reporting

Reporting Tools	ReportLab, Markdown-PDF	Automated generation of forensic reports
Virtualization Software	VMware Workstation 16	Running isolated test environments safely

8.3 Online Resources and APIs

Resource	Use Case
Google Drive API	Collecting and analyzing cloud file metadata
AWS S3 API	Retrieving and correlating object storage activity logs
Microsoft Graph API (OneDrive)	Accessing and verifying cloud access patterns
YARA Rule Repositories	Using open-source threat detection rules for ransomware indicators

8.4 Human and Support Resources

Resource	Description
Supervisor / Advisor	Provides technical and research guidance throughout the project.
Academic References & Online Documentation	Guides for Volatility, YARA, and forensic best practices.
Peer Feedback / Testing Assistance	Helps in cross-verifying forensic results and report validation.

Expected Outcome and Significance

9.1 Expected Outcomes

The **ForensicX** project is expected to deliver a fully functional prototype that automates the process of collecting, correlating, and reporting digital forensic evidence from multiple data sources. The system outcomes will demonstrate how a unified approach can reduce fragmentation in forensic investigations and improve accuracy in ransomware and cloud-based attack analysis.

Outcome	Description / Expected Result
Modular Forensic Framework	A Python-based prototype capable of integrating memory, log, and cloud data analysis within a single workflow.
Ransomware Detection and Correlation	Automated identification of ransomware indicators (entropy spikes, file renaming, IOCs) and timeline reconstruction.
Cross-Source Correlation Engine	Unified linkage between artifacts from different data sources to build a complete attack timeline.
Automated Forensic Report Generator	Generates PDF/HTML reports containing hashes, timestamps, evidence trails, and chain-of-custody details.
Secure Virtual Testing Environment	Validated framework performance through synthetic ransomware and cloud simulation in isolated VMs.

9.2 Significance of the Project

Domain	Significance / Contribution
Academic Contribution	ForensicX provides a reproducible research framework that bridges memory, log, and cloud forensics. It contributes to the field of digital forensics education and supports future academic experiments and teaching demonstrations.
Industrial Relevance	Offers a foundation for developing lightweight, open-source forensic tools suitable for small SOC (Security Operations Center) environments. It reduces investigation time and enhances evidence reliability.

Societal Impact	This contributes to faster and more accurate forensic responses against ransomware and cloud-targeted attacks, helping organizations and users minimize operational losses.
Research Advancement	Encourages the exploration of unified forensic systems and serves as a base for future enhancements such as AI-assisted detection or real-time analysis.

9.3 Long-Term Vision

In the long term, ForensicX can evolve into a scalable open-source platform that integrates additional features such as network forensics, blockchain-based evidence verification, and AI-driven threat classification. The system aims to support both academic learning and industry adaptation for modern cybersecurity challenges.

References

- Cohen, M., Bilby, D. & Caronni, G. (2011) *GRR Rapid Response: Forensic collection for remote live forensics*. Google Inc.
- Elastic NV (2021) *Elastic Security: Unified threat detection and response*. Elastic.co. Available at: https://www.elastic.co/security
- Google (2019) *Timesketch: Collaborative forensic timeline analysis*. Available at: https://github.com/google/timesketch
- Magnet Forensics (2017) *Magnet AXIOM Digital Forensics Suite*. Magnet Forensics. Available at: https://www.magnetforensics.com/
- The Volatility Foundation (2021) *Volatility 3 Framework: Advanced memory forensics*. Available at: https://github.com/volatilityfoundation/volatility3
- Velociraptor (2023) *Velociraptor: Open-source endpoint visibility and forensic collection*. Available at: https://www.velocidex.com/velociraptor/
- VirusTotal (2020) *YARA: The pattern matching Swiss knife for malware researchers*. VirusTotal. Available at: https://virustotal.github.io/yara/
- Casey, E. (2019) *Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet*. 4th edn. Academic Press.
- Alenezi, A. (2022) *A Comprehensive Review on Ransomware Detection and Forensic Techniques*. *Journal of Cyber Security Technology,* 6(3), pp. 145–162.
- Beebe, N. & Clarke, N. (2021) *Digital Forensics: Advances in Research and Practice*. Springer.
- Stallings, W. (2020) *Network Security Essentials: Applications and Standards*. 7th edn. Pearson Education.